Phase 2: Detailed Error Analysis Report

Team TLDR

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Introduction

We analyzed over **7,500 failed trajectories** across 9,900 tasks from the comprehensive dataset. The analysis reveals that the vast majority are still generic execution errors (logic violations) or the agent giving up early. These failures perfectly align with the 44 specific failure categories we isolated.

Quantitative Evaluation

To validate the severity of these failures, we quantitatively isolated the agent's baseline reliability and tracked the volume of distinct failure states.

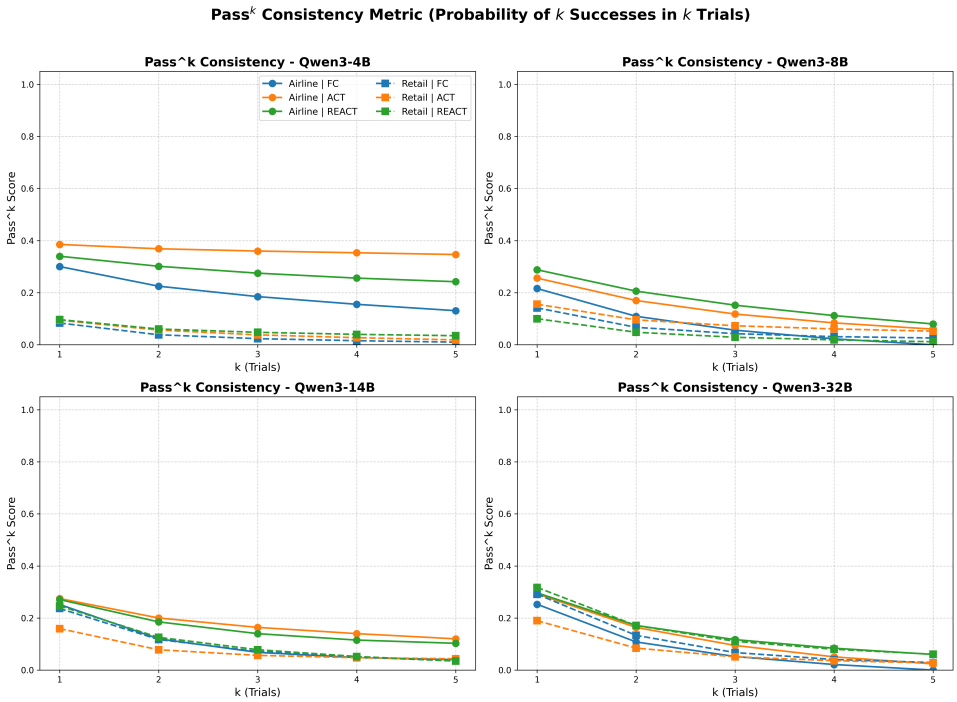

Figure 1: **Pass^k Consistency Probability across 4 Models.** This metric strictly calculates the probability of the agent achieving k successes out of k trials. As k moves from 1 to 5, the probability of consistent consecutive success exponentially drops near zero. This explicitly demonstrates that while the agent might stumble into success on a single attempt (Pass^1), it cannot reliably and consistently complete tasks without systemic guardrails.

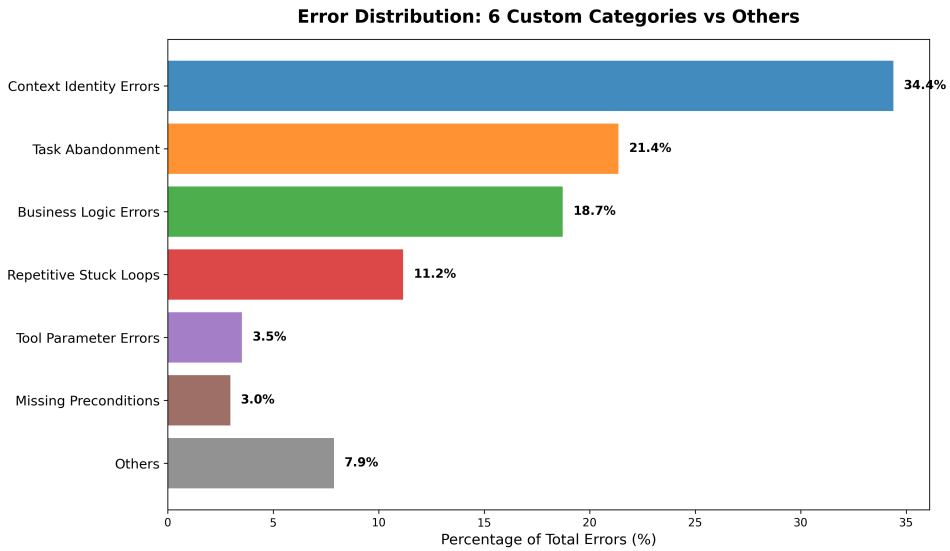

Figure 2: **Failure Distribution Mapped to Custom Taxonomy.** Upon extracting the explicit sub-failures across the 120 trajectory splits, we mapped the raw logs to our 6 formalized error buckets. The chart validates that our 6 custom categories mathematically capture around **90%** of all distinct failure loops generated by the agent. By solving just these 6 categorical buckets through a multi-agent hierarchy, the framework can eradicate nearly 90% of system failures.

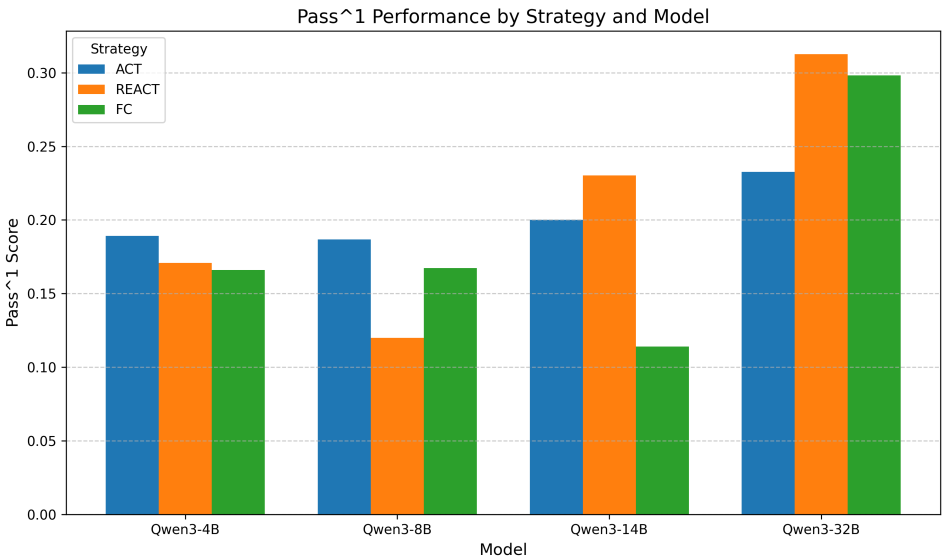

Figure 3: **Prompting Strategy Comparison (Act vs. ReAct vs. FC).** This chart evaluates the baseline success rates across three distinct tool-calling methodologies. The data explicitly demonstrates that simply altering the prompting strategy, whether forcing immediate action (Act), injecting sequential reasoning traces (ReAct), or relying on native json schemas (FC), is insufficient to overcome the fundamental bottlenecks. Even the most advanced reasoning instructions succumb to the same repetitive loop failures and logic errors without systemic, external architectural guardrails.

Tool Parameter Errors

The agent attempted to call tools with fabricated, malformed, or contextually incorrect parameters, representing strict interface interaction failures.

- **Acted without looking up first** – Agent called a write tool (exchange, modify, return) using item IDs it never actually retrieved from the database. The IDs were either fabricated or reused from memory.

  Metadata: Domain: Retail | Model: Qwen3-14B | Strategy: FC | Trial: 2 | Task ID: 86

- **Dropped the `#W` prefix on order ID** – Agent called `get_order_details` with `"order_id": "W3561391"` instead of `"#W3561391"`. The hash-W prefix is required format; omitting it causes an "order not found" error.

  Metadata: Domain: Retail | Model: Qwen3-14B | Strategy: FC | Trial: 1 | Task ID: 87

- **Used a product ID where a variant/item ID was needed** – The product ID (e.g. `9743693396`) identifies a product family. The item ID (e.g. `9879255677`) identifies a specific variant (size, color). Agent confused the two levels.

  Metadata: Domain: Retail | Model: Qwen3-4B | Strategy: REACT | Trial: 3 | Task ID: 72

- **Booked an airline flight without searching for availability first** – Agent called `book_reservation` or `update_reservation_flights` using a flight number from the existing reservation or from memory, without first running `search_direct_flight` to verify availability.

  Metadata: Domain: Airline | Model: Qwen3-4B | Strategy: ACT | Trial: 4 | Task ID: 46

- **Looked up a fake placeholder order ID** – Agent called `get_order_details` with `#W0000000` — a dummy ID seen in training examples — or with an order ID that doesn't appear in the user's profile returned by `get_user_details`.

  Metadata: Domain: Retail | Model: Qwen3-4B | Strategy: FC | Trial: 2 | Task ID: 6

- **Used a wrong or malformed payment method ID** – Agent attempted to refund to PayPal (policy violation) instead of a credit card. After the API error, it failed to generate a corrected JSON tool call and responded conversationally.

  Metadata: Domain: Retail | Model: Qwen3-32B | Strategy: ACT | Trial: 0 | Task ID: 13

Missing Preconditions

The agent failed to satisfy the required operational preconditions before executing actions, such as bypassing mandatory search steps or using incompatible arguments.

- **Passed argument names the tool doesn't accept** – Agent called a tool with keyword arguments that don't exist in that tool's schema, causing Python's `invoke()` to throw a `TypeError`.

  Metadata: Domain: Retail | Model: Qwen3-8B | Strategy: REACT | Trial: 3 | Task ID: 83

- **Called a tool that doesn't exist in the environment** – Agent invoked a tool name (action) that doesn't appear in the available tool set — either hallucinated from training data or reasoned to exist.

  Metadata: Domain: Airline | Model: Qwen3-4B | Strategy: ACT | Trial: 4 | Task ID: 45

- **Used wrong ID type for a flight lookup (airline)** – Agent passed the reservation ID where a flight number was required, or vice versa. Both are 6-character alphanumeric strings, so they look similar.

  Metadata: Domain: Airline | Model: Qwen3-32B | Strategy: FC | Trial: 2 | Task ID: 36

- **Forgot to include all passengers when booking** – A round-trip or multi-passenger reservation requires all passengers in every update/booking call. Agent included only some of them.

  Metadata: Domain: Airline | Model: Qwen3-8B | Strategy: FC | Trial: 2 | Task ID: 33

- **Items swapped in don't match items swapped out** – `exchange_delivered_order_items` requires `len(item_ids) == len(new_item_ids)`. Agent passed 2 old items but only 1 new item (or 1 old, 2 new).

  Metadata: Domain: Retail | Model: Qwen3-4B | Strategy: ACT | Trial: 4 | Task ID: 98

- **Used a travel certificate for a reservation update** – Certificates can only pay for new bookings, not for updating existing reservations. Agent used `certificate_200` as payment for `update_reservation_flights`.

  Metadata: Domain: Airline | Model: Qwen3-4B | Strategy: FC | Trial: 3 | Task ID: 33

Context Identity Errors

The agent incorrectly managed state context or user constraints, including processing invalid items, failing to match out-of-bounds inputs, or attempting repeated writes.

- **Booked a flight with insufficient seats** – Agent found a flight via search (appeared available), but during booking it had no remaining seats for the required number of passengers.

  Metadata: Domain: Airline | Model: Qwen3-14B | Strategy: FC | Trial: 1 | Task ID: 3

- **Submitted only one leg when updating a round-trip** – A round-trip reservation requires both legs (outbound + return) to be specified in every update call, even if only one is changing. Agent passed only the leg it was modifying.

  Metadata: Domain: Airline | Model: Qwen3-8B | Strategy: FC | Trial: 1 | Task ID: 41

- **Called a write tool more than once on the same order** – The agent correctly determined the variant count (10) but modified each pending order item separately using two `modify_pending_order_items` calls instead of a single consolidated action.

  Metadata: Domain: Retail | Model: Qwen3-32B | Strategy: REACT | Trial: 1 | Task ID: 4

- **Did an irreversible action then pretended to undo it** – Agent returned or cancelled something (irreversible), then subsequently tried to exchange or modify the same item/reservation as if the cancellation hadn't happened.

  Metadata: Domain: Retail | Model: Qwen3-8B | Strategy: REACT | Trial: 3 | Task ID: 93

- **Refunded to a different payment type than the original** – User's order was paid by PayPal, but agent refunded to a credit card (or vice versa). The tool accepts this silently; the evaluator catches it. Agent never notices.

  Metadata: Domain: Airline | Model: Qwen3-14B | Strategy: ACT | Trial: 0 | Task ID: 9

- **Missing Explicit Confirmation Before Exchange** – Agent failed to explicitly list both exchanges and failed to obtain user confirmation before calling the exchange tool, resulting in repeated conversational loops and premature termination.

  Metadata: Domain: Retail | Model: Qwen3-32B | Strategy: FC | Trial: 4 | Task ID: 58

Task Abandonment

The agent prematurely hallucinated task completion, imported out-of-session context, or abandoned the explicit guidelines established by the prompt.

- **Cost exceeded budget without offering alternatives** – Agent proceeded with an upgrade costing $1200 despite user's $600 limit; did not suggest upgrading the companion instead.

  Metadata: Domain: Airline | Model: Qwen3-4B | Strategy: ACT | Trial: 2 | Task ID: 43

- **Agent completed the task but never closed it** – The agent successfully found the correct flights and explicitly asked the user to confirm, but halted its execution entirely without invoking the final booking tool after the user gave the confirmation.

  Metadata: Domain: Airline | Model: Qwen3-8B | Strategy: ACT | Trial: 0 | Task ID: 0

- **Brought in context from somewhere else** – In its internal `<think>` block, the agent mentions a user name, email, or user ID that the user never provided in the conversation — imported from training data or a previous session.

  Metadata: Domain: Airline | Model: Qwen3-4B | Strategy: FC | Trial: 3 | Task ID: 11

- **Imported user identity from outside the session** – Agent's `<think>` block references an email address or user ID (e.g. `olivia_gonzalez_2305`) that the current user never mentioned — pulled from training or a different task.

  Metadata: Domain: Retail | Model: Qwen3-4B | Strategy: FC | Trial: 3 | Task ID: 8

- **Used placeholder credentials instead of asking** – The agent called a tool with a dummy placeholder string like `"user_id_needed"`. When this threw an error, the agent panicked and escalated to a human agent instead of asking the user for their real info.

  Metadata: Domain: Airline | Model: Qwen3-14B | Strategy: REACT | Trial: 0 | Task ID: 0

- **Authentication Failure and Unnecessary Human Transfer** – Agent used incorrect authentication details (lucas.johnson@example.com), failed the user lookup, and prematurely transferred to a human agent despite the request being fully solvable with available tools.

  Metadata: Domain: Retail | Model: Qwen3-32B | Strategy: REACT | Trial: 2 | Task ID: 104

Business Logic Errors

The agent failed to respect core domain logic, committing errors such as soliciting sensitive payment data, promising non-existent benefits, or ignoring strict constraints.

- **Kept asking for information it could have looked up** – Agent asked the user for their order ID, tracking number, or item name 3+ times across the conversation when it could have retrieved this from `get_user_details` or `get_order_details`.

  Metadata: Domain: Retail | Model: Qwen3-4B | Strategy: FC | Trial: 1 | Task ID: 74

- **Said "I'll do X" but never did X** – Agent verbally committed in its text response to performing an action (return, cancel, exchange, transfer), but no corresponding tool call was made before the conversation ended.

  Metadata: Domain: Retail | Model: Qwen3-8B | Strategy: ACT | Trial: 4 | Task ID: 72

- **Told the user it succeeded when the tool returned an error** – The agent falsely claimed to the user that the task was complete and hallucinated a fake confirmation code, violating the instruction by never actually invoking the final API tool correctly.

  Metadata: Domain: Airline | Model: Qwen3-8B | Strategy: FC | Trial: 0 | Task ID: 0

- **Asked user for credit card number or CVV** – Agent asked the user to provide raw payment credentials in the chat — information the agent cannot process and which violates PCI rules.

  Metadata: Domain: Airline | Model: Qwen3-8B | Strategy: REACT | Trial: 3 | Task ID: 26

- **Offered benefits the system cannot provide** – Agent verbally committed to offering fee waivers, filing DOT complaints, or sending gift cards — none of which it has tools to do.

  Metadata: Domain: Airline | Model: Qwen3-32B | Strategy: FC | Trial: 4 | Task ID: 17

- **Used a gift card without checking if the balance covers the cost** – Agent attempted a transaction using a gift card as payment, without first comparing the card's balance to the action's cost. Discovered only when the API returned an insufficient balance error.

  Metadata: Domain: Retail | Model: Qwen3-14B | Strategy: ACT | Trial: 2 | Task ID: 50

Repetitive Stuck Loops

The agent repeatedly executed the same failed strategy, processed unrequested items, or got stuck mirroring invalid states without forward progress.

- **Premature Task Termination** – Agent failed to properly confirm all exchange details and ended the conversation with `###STOP###` without issuing the required final exchange tool call.

  Metadata: Domain: Retail | Model: Qwen3-32B | Strategy: ACT | Trial: 2 | Task ID: 1

- **Acted on the wrong order (item wasn't in it)** – User described a specific product to return/exchange. Agent retrieved an order that doesn't contain that product and called write tools on it anyway.

  Metadata: Domain: Retail | Model: Qwen3-8B | Strategy: FC | Trial: 3 | Task ID: 70

- **Returned more items than the user asked for** – User said "return one item." Agent included 2+ item IDs in the return call — processing an unrequested return on a different item in the same order.

  Metadata: Domain: Retail | Model: Qwen3-14B | Strategy: ACT | Trial: 0 | Task ID: 112

- **Retrieved the wrong order data and proceeded anyway** – User asked about a specific product (shoes, camera, etc.). Agent retrieved an order that doesn't contain that product, then called write tools on that order anyway without noticing the mismatch.

  Metadata: Domain: Retail | Model: Qwen3-8B | Strategy: ACT | Trial: 3 | Task ID: 72

- **Made a permanent address change with a city the user never mentioned** – Agent called `modify_user_address` with a city not mentioned anywhere in the conversation, without asking what the new address should be.

  Metadata: Domain: Airline | Model: Qwen3-14B | Strategy: FC | Trial: 3 | Task ID: 23

- **User sent a non-English message** – User communicated in French, German, or Spanish. Agent either failed to extract identity information from the non-English text, used placeholder values it inferred, or transferred immediately.

  Metadata: Domain: Retail | Model: Qwen3-8B | Strategy: REACT | Trial: 2 | Task ID: 64

Other unexpected behaviors

The following errors represent less common edge cases that still cause significant pipeline disruption. They typically involve edge-case context breakdown or premature loop termination.

- **Unconfirmed Database Writes**

  Metadata: Domain: Airline | Model: Qwen3-32B | Strategy: FC | Trial: 3 | Task ID: 6

- **Gave up before trying anything**

  Metadata: Domain: Retail | Model: Qwen3-8B | Strategy: ACT | Trial: 0 | Task ID: 0

- **Gave up after one failed step**

  Metadata: Domain: Retail | Model: Qwen3-4B | Strategy: ACT | Trial: 3 | Task ID: 73

- **Stopped after first non-matching order**

  Metadata: Domain: Retail | Model: Qwen3-14B | Strategy: REACT | Trial: 2 | Task ID: 41

- **Sent user to an external channel instead of using tools**

  Metadata: Domain: Retail | Model: Qwen3-14B | Strategy: FC | Trial: 3 | Task ID: 44

- **Got stuck calling the same tool repeatedly**

  Metadata: Domain: Airline | Model: Qwen3-8B | Strategy: REACT | Trial: 0 | Task ID: 0

- **Agent and user mirrored each other until step limit**

  Metadata: Domain: Airline | Model: Qwen3-32B | Strategy: ACT | Trial: 4 | Task ID: 27

- **Echo Mirror Loop**

  Metadata: Domain: Airline | Model: Qwen3-14B | Strategy: ACT | Trial: 1 | Task ID: 5

Individual Contributions

The workload was distributed among the five team members as follows:

- **Harshith (Qwen3-4B):** Evaluated error categories and extracted relevant examples from Qwen3-4B-generated JSON files.

- **Shravan (Qwen3-8B):** Evaluated error categories and extracted examples from Qwen3-8B-generated JSON files.

- **Gouri (Qwen3-14B):** Evaluated error categories and extracted examples from Qwen3-14B-generated JSON files.
- **Vidya (Qwen3-32B - Retail):** Evaluated error categories and extracted examples from Qwen3-32B-Retail-generated JSON files.
- **Vedang (Qwen3-32B - Airline):** Evaluated error categories and extracted examples from Qwen3-32B-Airline-generated JSON files.
- **Equal Contribution:** Categorized the failed JSON tasks into their respective error categories.

Additional Resources

1) All 120 JSON experimental files, datasets, and detailed logs are available in the DropBox link provided below: Click here for Phase 1's DropBox link

2) The JSON examples of the error categories that we extracted and categorized are available in the DropBox link provided below: Click here for Phase 2's DropBox link